Parasympatholytics (Anticholinergic drugs)

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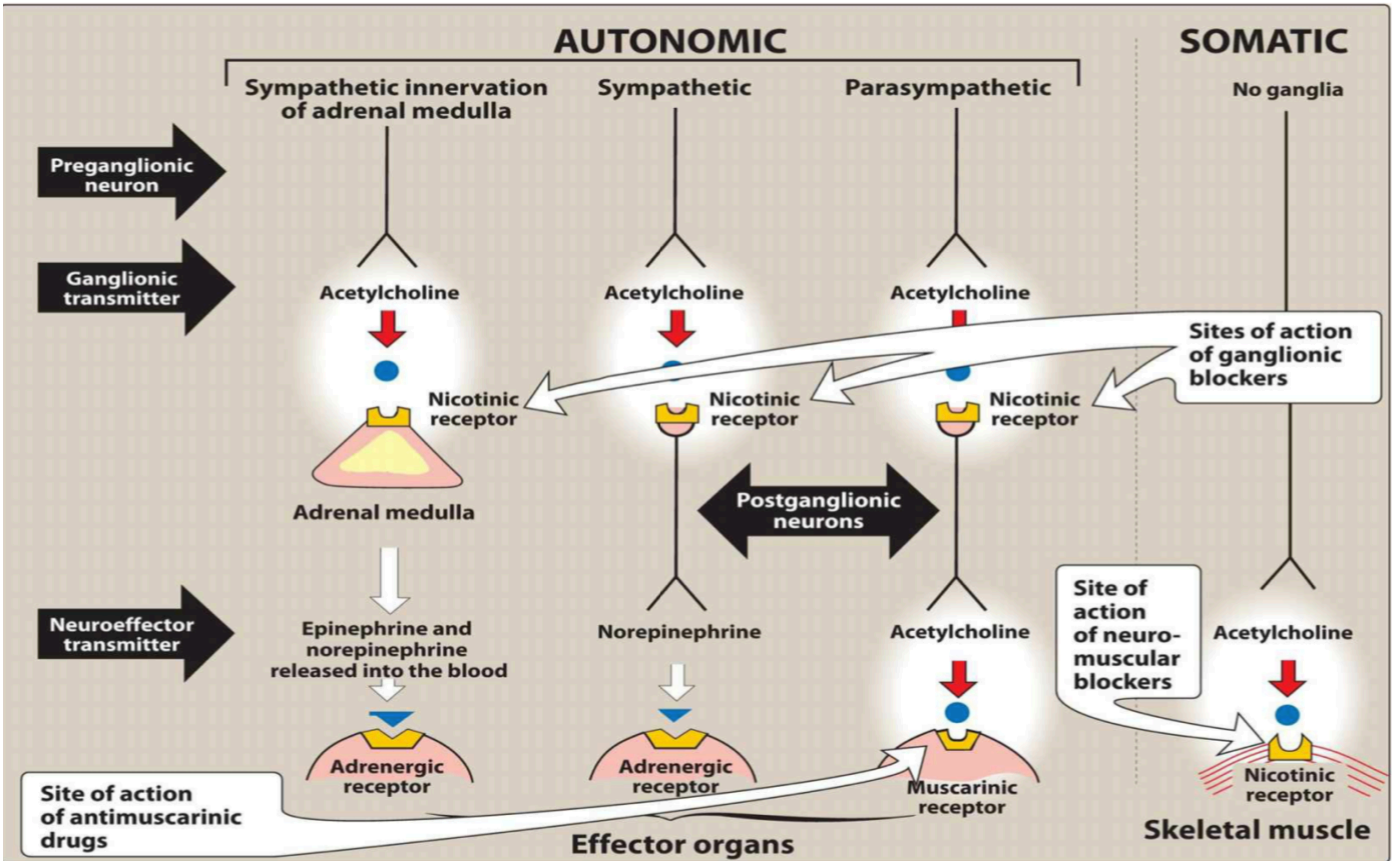
Objectives

- Know the names and mechanism of action of anticholinergic drugs
- List the indications and side effects of these drugs

Muscarinic antagonists (Antimuscarinic Agents)

- Commonly known as anticholinergic drugs, these agents (for example, atropine and scopolamine) block muscarinic receptors, causing inhibition of muscarinic functions.
- In addition, these drugs block the few exceptional sympathetic neurons that are cholinergic, such as those innervating the salivary and sweat glands.

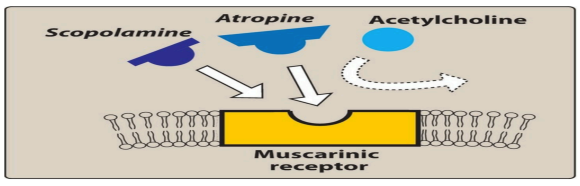
- Because they do not block nicotinic receptors, the anticholinergic drugs (more precisely, antimuscarinic drugs) have little or no action at skeletal neuromuscular junctions (NMJs) or autonomic ganglia.
- The anticholinergic drugs are beneficial in a variety of clinical situations.



A. Atropine



- Atropine is a tertiary amine extracted from belladonna alkaloid.
- It has a high affinity for muscarinic receptors and binds competitively to prevent ACh from binding.
- Atropine acts both centrally and peripherally.
- General actions last about 4 hours; however, effects of topical administration in the eye may persist for days.
- Neuroeffector organs have varying sensitivity to atropine. The greatest inhibitory effects are seen in bronchial tissue, salivary and sweat glands, and the heart.



Actions

a. Eye

Atropine blocks muscarinic activity in the eye, resulting in mydriasis (dilation of the pupil), unresponsiveness to light, and cycloplegia (inability to focus for near vision).

In patients with angle-closure glaucoma, intraocular pressure may rise dangerously.



b. Gastrointestinal (GI)

- Atropine can be used as an antispasmodic to reduce activity of the GI tract.
- Atropine and scopolamine are probably the most potent antispasmodic drugs available.
- Although gastric motility is reduced, hydrochloric acid production is not significantly affected. Thus, atropine is not effective for the treatment of ulcers.
- Doses of atropine that reduce spasms also reduce saliva secretion, ocular accommodation, and urination. These effects decrease compliance with atropine.

c. Cardiovascular

- Atropine produces divergent effects on the cardiovascular system, depending on the dose.
- At low doses, the predominant effect is a slight decrease in heart rate. This effect results from blockade of M1 receptors on the inhibitory prejunctional (or presynaptic) neurons, thus permitting increased ACh release.
- Higher doses of atropine cause a progressive increase in heart rate by blocking M2 receptors on the sinoatrial node.

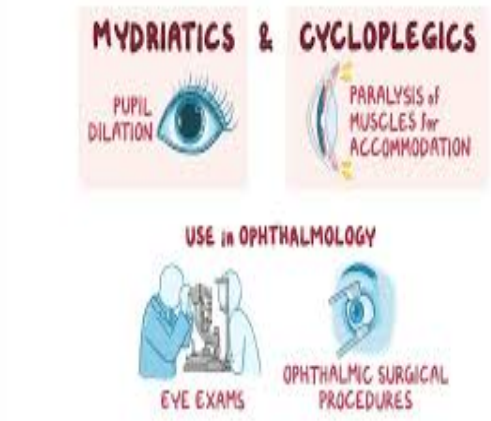
d. Secretions

- Atropine blocks muscarinic receptors in the salivary glands, producing dryness of the mouth (xerostomia).
- The salivary glands are exquisitely sensitive to atropine.
- Sweat and lacrimal glands are similarly affected. [Note: Inhibition of secretions of sweat glands can cause elevated body temperature, which can be dangerous in children and the elderly.]

Therapeutic uses

a. Ophthalmic

- Topical atropine exerts both mydriatic and cycloplegic effects, and it permits the measurement of refractive errors without interference by the accommodative capacity of the eye.
- Shorter-acting antimuscarinics (cyclopentolate and tropicamide) have largely replaced atropine due to prolonged mydriasis observed with atropine (7 to 14 days vs. 6 to 24 hours with other agents).



b. Antispasmodic

Atropine is used as an antispasmodic agent to relax the GI tract.

c. Cardiovascular

Injectable atropine is used to treat bradycardia of varying etiologies.

d. Antisecretory

Atropine is sometimes used as an antisecretory agent to block secretions in the respiratory tract prior to surgery.

e. Antidote for cholinergic agonists

- Atropine is used for the treatment of organophosphate (insecticides, nerve gases) poisoning, of overdose of clinically used anticholinesterases such as physostigmine, and in some types of mushroom poisoning (certain mushrooms contain cholinergic substances that block cholinesterases).
- Massive doses of injectable atropine may be required over a long period to counteract the poisons.
- The ability of atropine to enter the central nervous system (CNS) is of particular importance in treating central toxic effects of anticholinesterases.

Pharmacokinetics

- Atropine is readily absorbed, partially metabolized by the liver, and eliminated primarily in urine.
- It has a half-life of about 4 hours.

Adverse effects

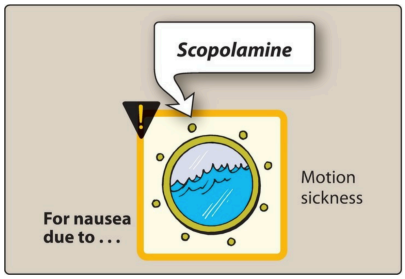
- Depending on the dose, atropine may cause dry mouth, blurred vision, "sandy eyes," tachycardia, urinary retention, and constipation.
- Effects on the CNS include restlessness, confusion, hallucinations, and delirium, which may progress to depression, collapse of the circulatory and respiratory systems, and death.
- Low doses of cholinesterase inhibitors, such as physostigmine, may be used to overcome atropine toxicity.
- Atropine may also induce troublesome urinary retention.
- The drug may be dangerous in children, because they are sensitive to its effects, particularly to rapid increases in body temperature.

B. Scopolamine

- Scopolamine, another tertiary amine plant alkaloid, produces peripheral effects similar to those of atropine.
- However, scopolamine has greater action on the CNS (unlike atropine, CNS effects are observed at therapeutic doses) and a longer duration of action as compared to atropine.

Actions

- Scopolamine is one of the most effective drugs available for motion sickness.
- It also has the unusual effect of blocking short-term memory.
- In contrast to atropine, scopolamine produces sedation, but at higher doses, it can produce excitement.
- Scopolamine may produce euphoria and is susceptible to abuse.



Therapeutic uses

- Scopolamine is used for the prevention of motion sickness and postoperative nausea and vomiting. For motion sickness, it is available as a topical patch that provides effects for up to 3 days.

Pharmacokinetics and adverse effects

- These aspects are similar to those of atropine, with the exception of longer half-life.

C. Ipratropium, and tiotropium

- Ipratropium and tiotropium are quaternary derivatives of atropine.
- These agents are approved as bronchodilators for maintenance treatment of bronchospasm associated with chronic obstructive pulmonary disease (COPD).
- Ipratropium and tiotropium are also used in the acute management of bronchospasm in asthma and chronic management of asthma, respectively.
- All of these agents are delivered via inhalation.
- Because of the positive charge, these drugs do not enter the systemic circulation or the CNS, restricting effects to the pulmonary system.

D. Tropicamide and cyclopentolate

- These agents are used as ophthalmic solutions for mydriasis and cycloplegia.
- Their duration of action is shorter than that of atropine.
- Tropicamide produces mydriasis for 6 hours and cyclopentolate for 24 hours.

E. Benztropine and trihexyphenidyl

- Benztropine and trihexyphenidyl are useful as adjuncts with other antiparkinson agents to treat Parkinson disease and other types of parkinsonian syndromes, including antipsychotic-induced extrapyramidal symptoms.

F. Antimuscarinic agents for overactive bladder

- Oxybutynin and tolterodine are synthetic atropine-like drugs with antimuscarinic actions.

Actions

- By competitively blocking muscarinic (M3) receptors in the bladder, intravesical pressure is lowered, bladder capacity is increased, and the frequency of bladder contractions is reduced.
- Antimuscarinic actions at M3 receptors in the GI tract, salivary glands, CNS, and eye may cause adverse effects.

Therapeutic uses

- These agents are used for management of overactive bladder and urinary incontinence.
- Oxybutynin is also used in patients with neurogenic bladder.

Pharmacokinetics

- These agents are available in oral dosage forms.
- They have a long half-life, which allows once-daily administration.
- Oxybutynin is also available in a transdermal patch and topical gel formulation.
- These drugs are hepatically metabolized by the cytochrome P450 system.

Adverse effects

- Side effects include dry mouth, constipation, and blurred vision, which limit tolerability of these agents.
- Extended- release formulations and the transdermal patch have a lower incidence of adverse effects and may be better tolerated.

Ganglionic Blockers

- Ganglionic blockers specifically act on the nicotinic receptors of both parasympathetic and sympathetic autonomic ganglia.
- Some also block the ion channels of the autonomic ganglia.
- These drugs show no selectivity toward the parasympathetic or sympathetic ganglia and are not effective as neuromuscular antagonists. Thus, these drugs block the entire output of the autonomic nervous system at the nicotinic receptor.
- Except for nicotine, the other drugs mentioned in this category are nondepolarizing, competitive antagonists.
- The responses of the nondepolarizing blockers are complex and mostly unpredictable. Therefore, ganglionic blockade is rarely used therapeutically, but often serves as a tool in experimental

Trimetaphan

- A short-acting agent, is given intravenously for the emergency control of hypertension.

Neuromuscular Blocking Agents

- These drugs block cholinergic transmission between motor nerve endings and the nicotinic receptors on skeletal muscle.
- They possess some chemical similarities to ACh and act either as antagonists (nondepolarizing) or as agonists (depolarizing) at the receptors on the endplate of the NMJ.
- Neuromuscular blockers (NMBs) are clinically useful to facilitate rapid intubation when needed due to respiratory failure (rapid sequence intubation).
- During surgery, they are used to facilitate endotracheal intubation and provide complete muscle relaxation at lower anesthetic doses. This increases the safety of anesthesia by allowing patients to recover quickly and completely.

References

- **Lippincott's Illustrated Reviews Pharmacology 7th edition.**